POPE 1351: observed visual reading vs causal instruction add-back
Image maps: head/layer/query means, visual-conditional. Zero E1 instruction map is imposed by mask.

Original: cyan=official target; orange=actual crop



Actual model input, 336x336
OFFICIAL_VISIBLE



```
Is there a truck in the image?
Gold=Yes; C3=Yes; E1=No

R position | region | native subword
581  Separ  _
582  Separ  [NL]
583  Q      Is
584  Q      _there
585  Q      _a
586  Q      _tr
587  Q      uck
588  Q      _in
589  Q      _the
590  Q      _image
591  Q      ?
592  Separ  [NL]
593  Instr  Answer
594  Instr  _the
595  Instr  _question
596  Instr  _using
597  Instr  _a
598  Instr  _single
599  Instr  _word
600  Instr  _or
601  Instr  _phrase
602  Instr  .
603  ASSIS  _A
604  ASSIS  SS
605  ASSIS  IST
606  ASSIS  ANT
```

C3 Question
Image=0.189, edge=33.3%



C3 Instruction
Image=0.104, edge=25.5%



C3 ASSISTANT
Image=0.167, edge=33.5%



E1 Question
Image=0.189, edge=33.3%

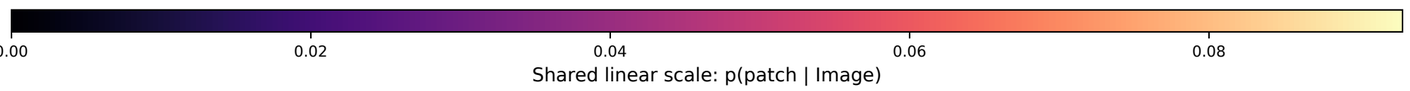


E1 Instruction
Image=0.000, edge=NA



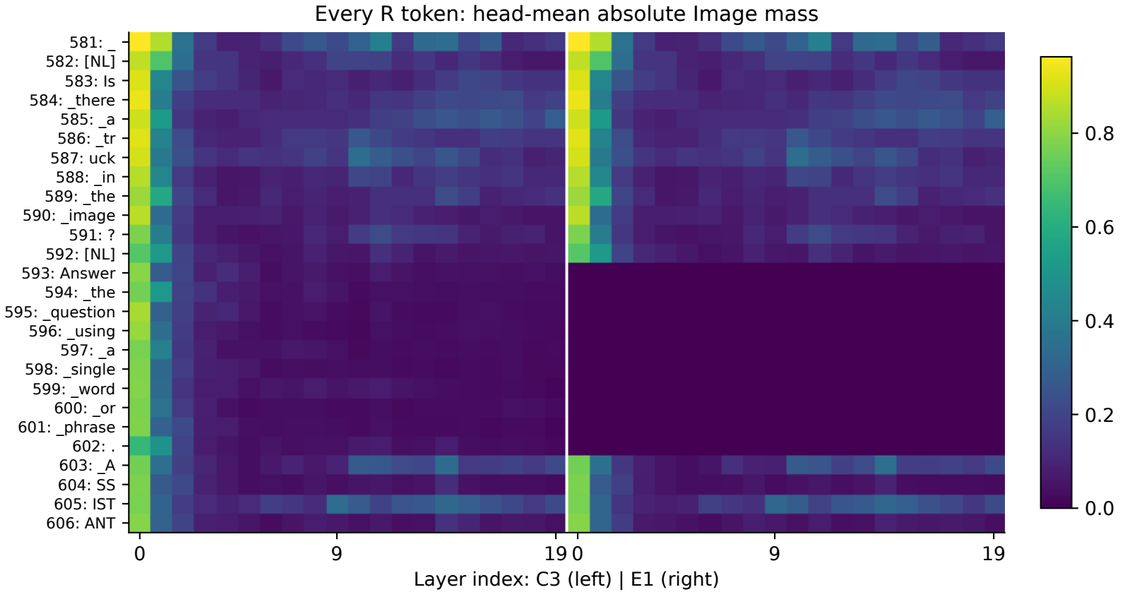
E1 ASSISTANT
Image=0.163, edge=33.1%





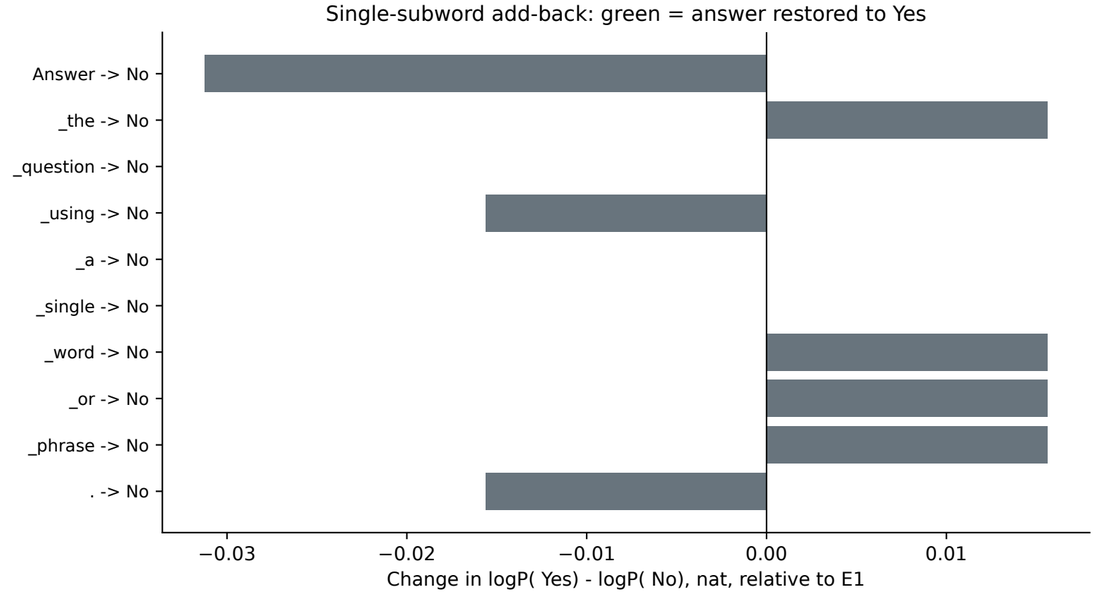
Shared linear scale: p(patch | Image)

Every R token: head-mean absolute Image mass

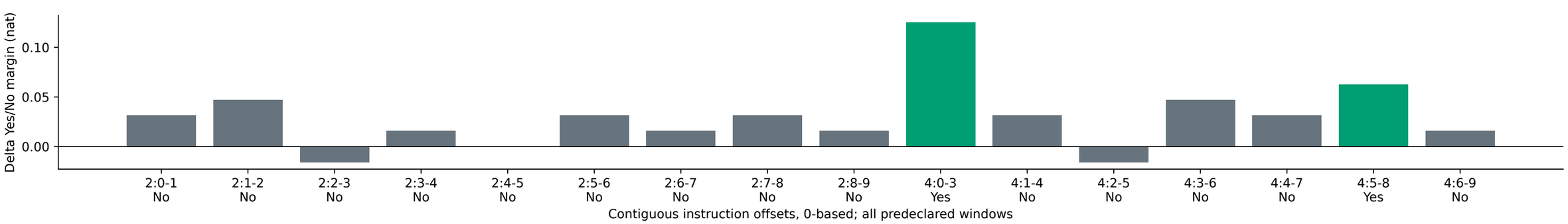


Layer index: C3 (left) | E1 (right)

Single-subword add-back: green = answer restored to Yes



Change in logP(Yes) - logP(No), nat, relative to E1



Contiguous instruction offsets, 0-based; all predeclared windows